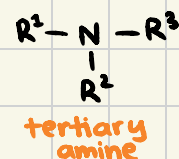
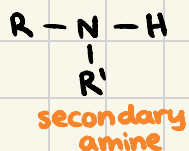
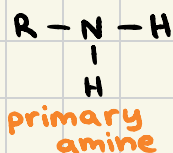
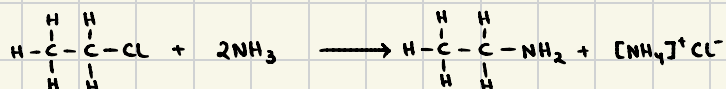


34 - Nitrogen compounds



Production of primary & secondary amines

1) halogenoalkane + ammonia (in ethanol)

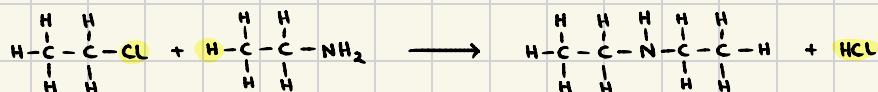


halogenoalkane + ammonia $\longrightarrow$ primary amine

reagent: NH_3 in ethanol

conditions: heat under pressure

2) halogenoalkane + primary amine (in ethanol)

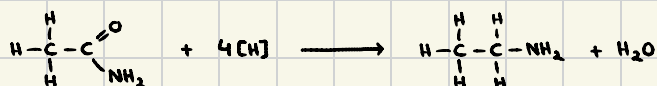


halogenoalkane + primary amine $\longrightarrow$ secondary amine

reagent: primary amine in ethanol

conditions: heat under pressure

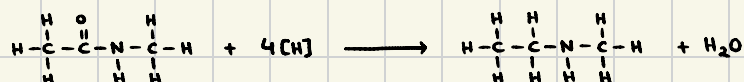
3) reduction of amides with LiAlH_4



primary amide + LiAlH_4 (reducing agent) $\longrightarrow$ primary amine

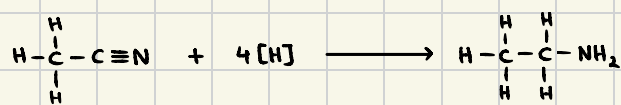
reagent: LiAlH_4 (reducing agent)

conditions: in dry ether

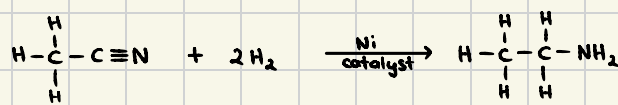


secondary amide + LiAlH_4 (reducing agent) $\longrightarrow$ secondary amine

4) reduction of nitriles with LiAlH_4 OR H_2 with Ni catalyst



OR

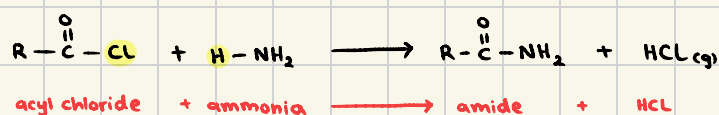


nitrile + LiAlH_4 $\longrightarrow$ amine
(in dry ether)

nitrile + H_2 $\longrightarrow$ amine

Production of amides

1) acyl chloride + ammonia

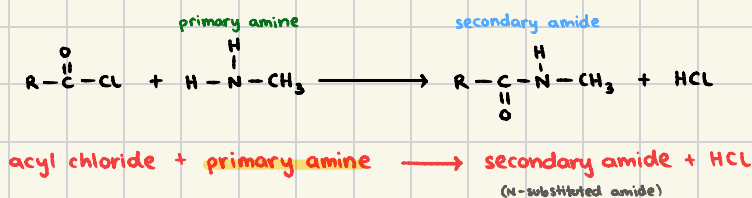


reagent: ammonia

conditions: room temp.

products: primary amide + HCl

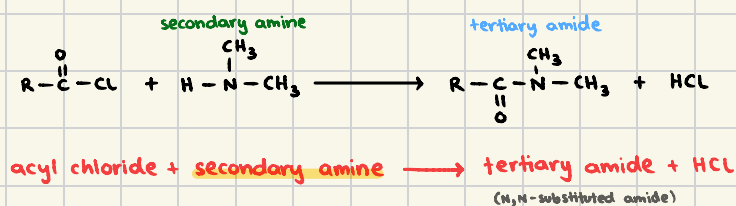
2) acyl chloride + amine



reagent: primary amine

conditions: room temp.

products: N-substituted amide



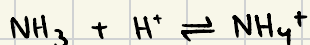
reagent: secondary amine

conditions: room temp.

products: N,N-substituted amide

Basicity of amines

Amines are WEAK BASES due to the lone pair of e^- on the nitrogen atom, which can accept H^+ in aq. solution



Primary, secondary & tertiary amines have different basicities because alkyl groups donate e^- density towards the nitrogen (inductive effect), enhancing the availability of the nitrogen lone pair for protonation

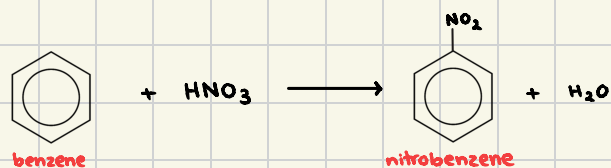
amines are more basic than ammonia due to alkyl groups increasing e^- density on nitrogen

aromatic amines are less basic because the lone pair on nitrogen is partially delocalised into the aromatic ring, making it less available for protonation

Preparation of phenylamine

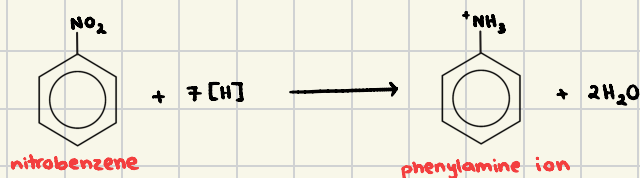
can be produced in a 3-step synthesis reaction

STEP 1: Nitration



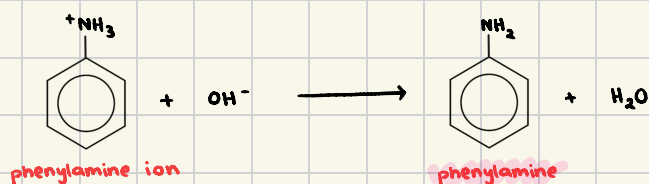
undergoes nitration with conc. HNO_3 & conc. H_2SO_4 at 25 to 60°C

STEP 2: Reduction



reduced with hot tin (sn) & conc. HCl under reflux

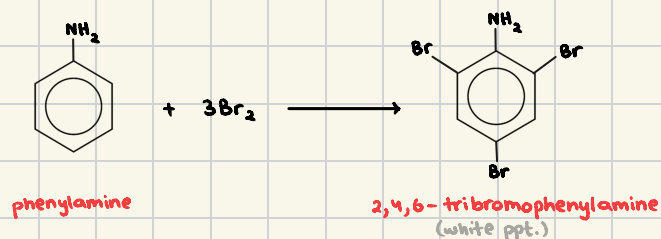
STEP 3: Deprotonation



NaOH is added to deprotonate the ion

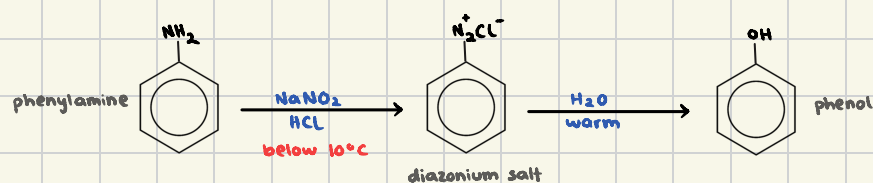
Reactions of phenylamine

1) reaction with $\text{Br}_2(\text{aq})$



reaction occurs at room temp.

2) formation of diazonium salt



Relative basicities: ethylamine VS aq ammonia VS phenylamine

strongest base

ethylamine



positive inductive effect, alkyl group donates e^- density to the nitrogen, causing its lone pair of e^- to become more available

aq ammonia



no e^- donating groups to cause positive inductive effect & no aromatic rings to cause delocalisation of nitrogen's lone pair of e^-

weakest base

phenylamine

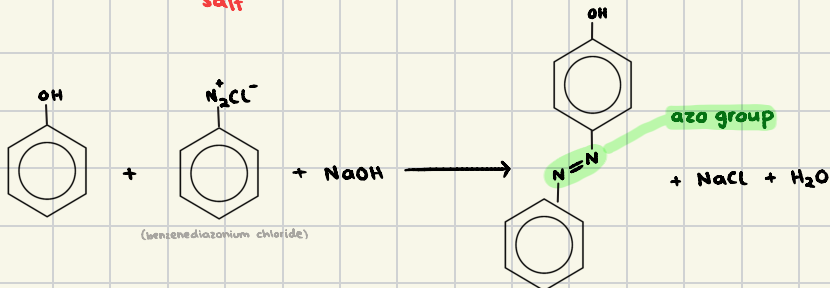


nitrogen's lone pair of e^- becomes delocalised in the benzene ring & is therefore LESS available to form a dative covalent bond with H^+

Azo compounds

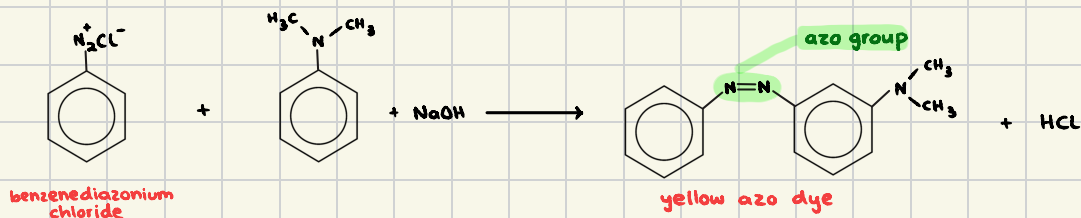
→ formation of azo compounds

phenol + diazonium salt + NaOH → azo compound



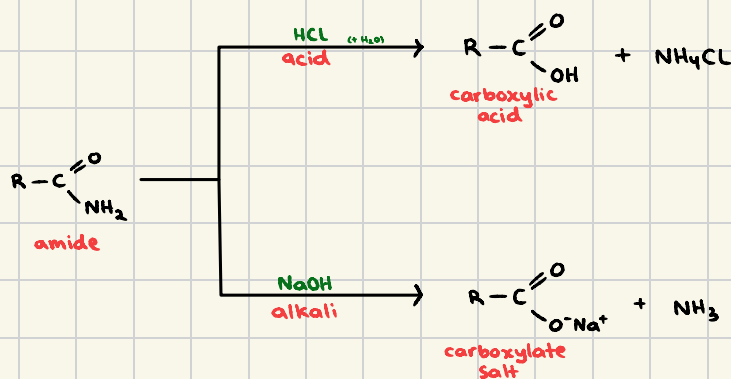
azo compounds are often used as dyes

you can make other azo dyes via similar routes

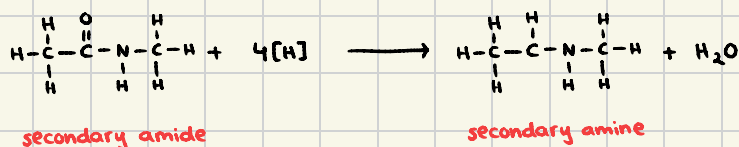
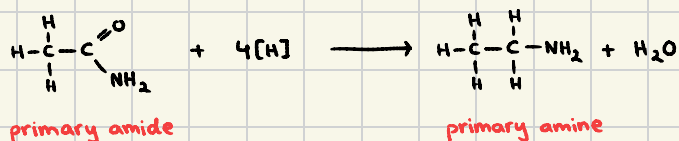


Reactions of amides

1) hydrolysis with aq. acid or alkali



2) reduction with LiAlH_4 (in dry ether)



Why are amides much weaker bases than amines?

① electron delocalisation

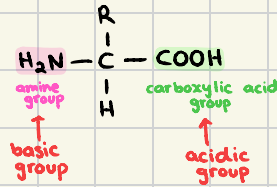
- ↳ in amides, the lone pair on the nitrogen is partially delocalised by the carbonyl group (resonance)
- ↳ this resonance reduces the availability of the nitrogen lone pair for protonation
- ↳ this makes amides less basic
- ↳ in amines, the nitrogen lone pair is localised & readily available for bonding with protons
- ↳ this makes amines more basic

② effect of resonance

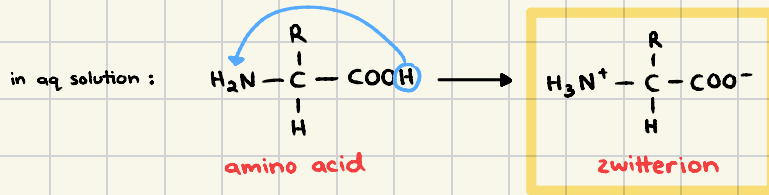
- ↳ the partial double-bond character in the C-N bond of amides (due to resonance), further stabilises the molecule
- ↳ this makes amides less basic

Amino acids

general structure of amino acids :



→ acid-base properties of amino acids



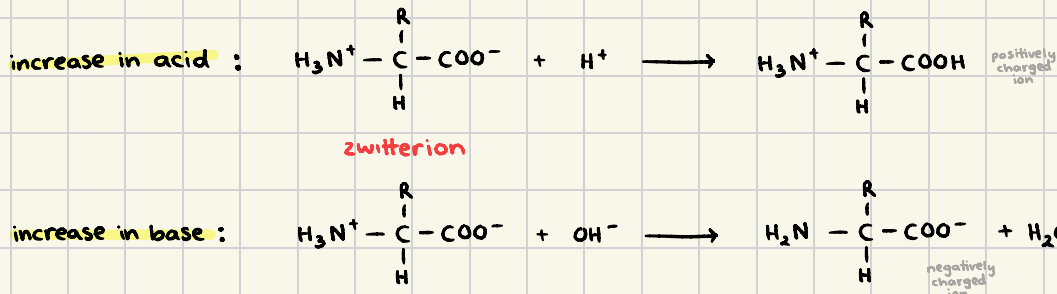
* in aq solutions, amino acids exhibit **amphoteric** behavior

* the NH_3^+ has the ability to donate protons

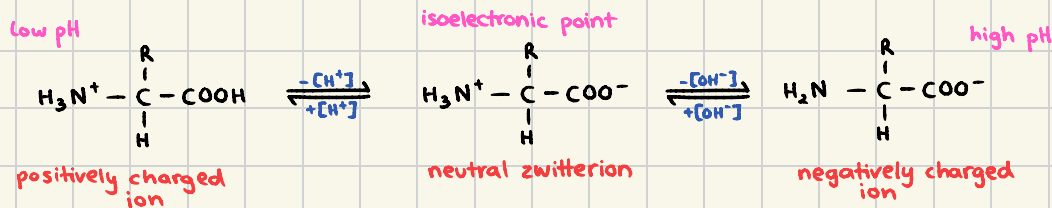
* the COO^- has the ability to accept protons

Zwitterion : a compound with no overall charge, but which contains separate parts which are positively & negatively charged

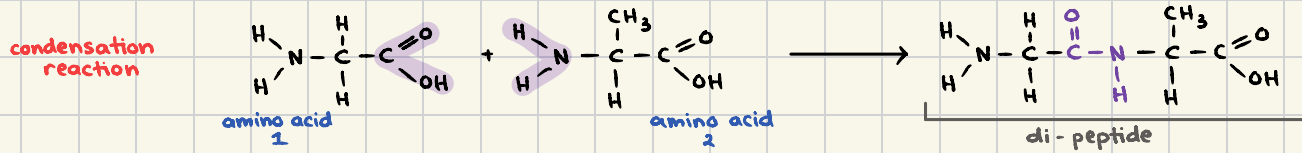
acts as a
buffer solution



The isoelectronic point of amino acids is the pH at which the amino acid exists as a zwitterion

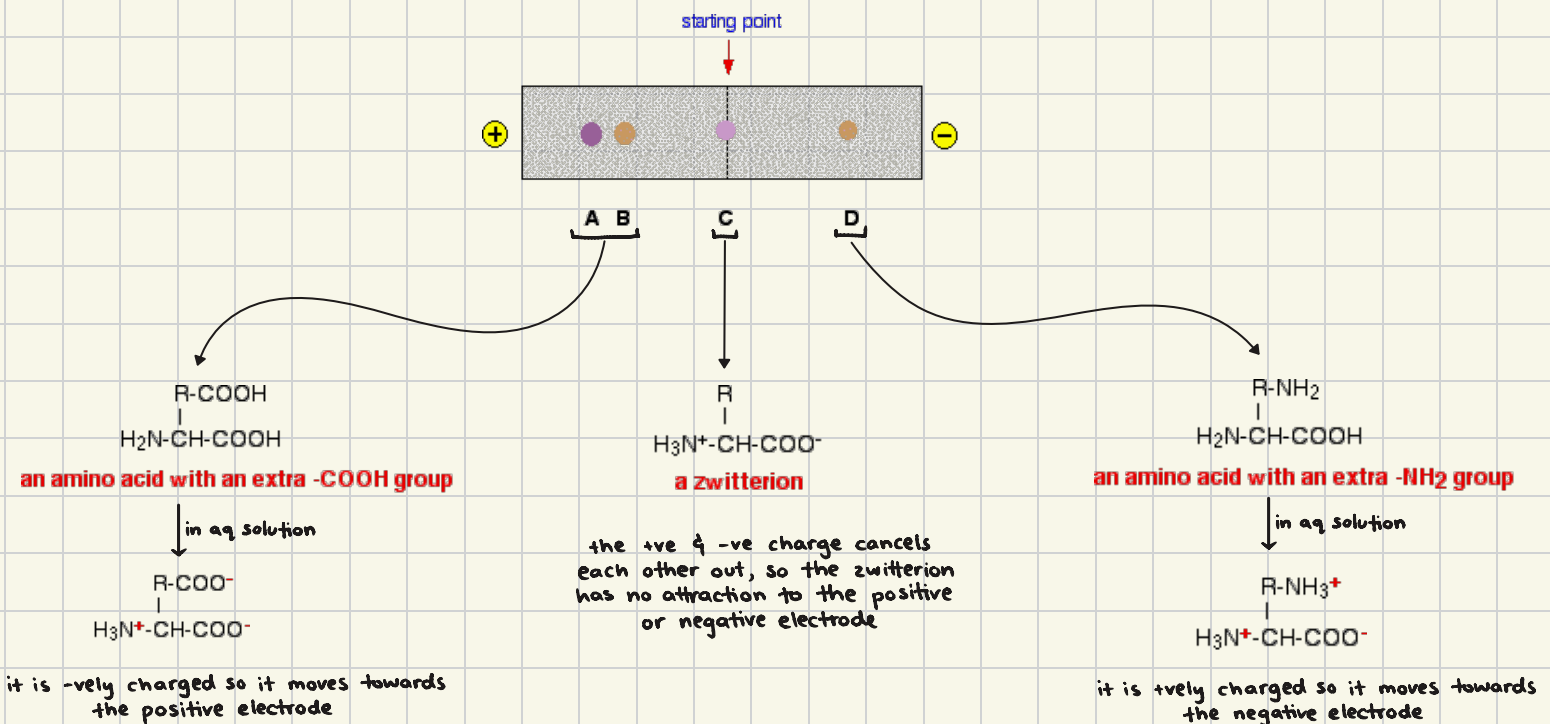
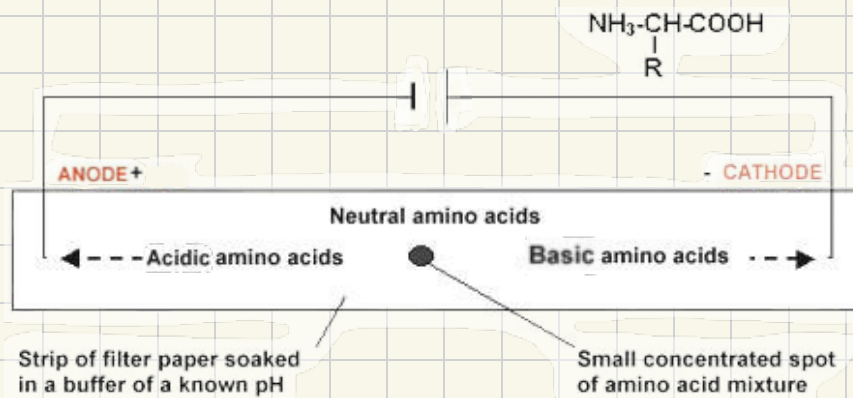


Formation of amide (peptide) bonds:



di-peptide : made up of two amino acids
 tri-peptide : made up of three amino acids
 polypeptide : made up of many amino acids

Electrophoresis



why did A & B travel different distances? ion size; smaller ions travel faster than bigger ones